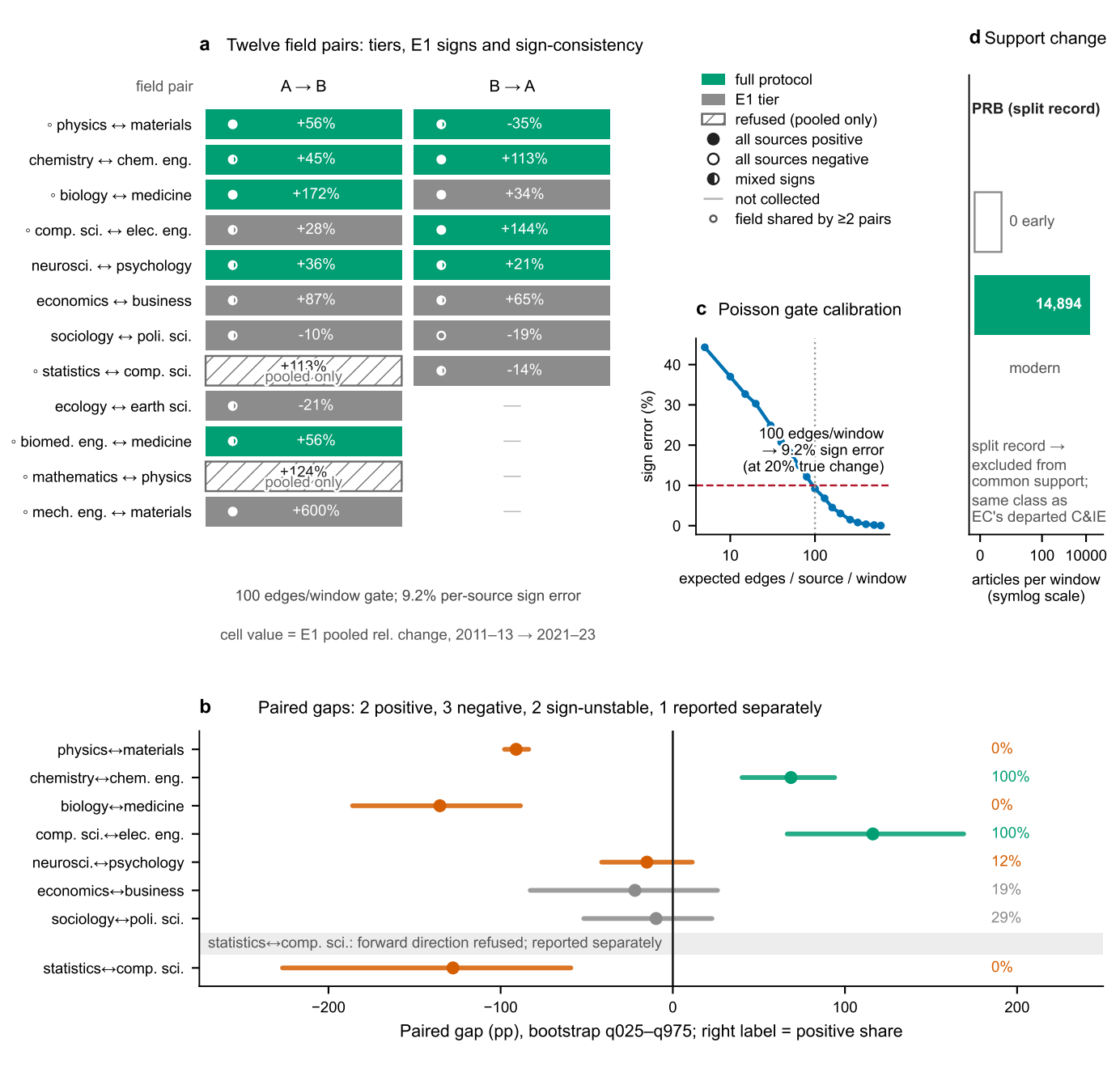

a Twelve field pairs: tiers, E1 signs and sign-consistency
field pair
A → B
B → A
◦ physics ↔ materials
+56%
-35%
chemistry ↔ chem. eng.
+45%
+113%
◦ biology ↔ medicine
+172%
+34%
◦ comp. sci. ↔ elec. eng.
+28%
+144%
neurosci. ↔ psychology
+36%
+21%
economics ↔ business
+87%
+65%
sociology ↔ poli. sci.
-10%
-19%
◦ statistics ↔ comp. sci.
+113%
pooled only
-14%
ecology ↔ earth sci.
-21%
◦ biomed. eng. ↔ medicine
+56%
◦ mathematics ↔ physics
+124%
pooled only
◦ mech. eng. ↔ materials
+600%
100 edges/window gate; 9.2% per-source sign error
cell value = E1 pooled rel. change, 2011–13 → 2021–23
full protocol
E1 tier
refused (pooled only)
all sources positive
all sources negative
mixed signs
not collected
field shared by ≥2 pairs
c Poisson gate calibration
sign error (%)
40
30
20
10
0
100 edges/window
→ 9.2% sign error
(at 20% true change)
10
100
expected edges / source / window
d Support change
PRB (split record)
0 early
14,894
modern
split record →
excluded from
common support;
same class as
EC's departed C&IE
0
100
10000
articles per window
(symlog scale)
b Paired gaps: 2 positive, 3 negative, 2 sign-unstable, 1 reported separately
physics↔materials
chemistry↔chem. eng.
biology↔medicine
comp. sci.↔elec. eng.
neurosci.↔psychology
economics↔business
sociology↔poli. sci.
statistics↔comp. sci.: forward direction refused; reported separately
statistics↔comp. sci.
0%
100%
0%
100%
12%
19%
29%
0%
−200
−100
0
100
200
Paired gap (pp), bootstrap q025–q975; right label = positive share